

Xania Hotel Booking Database — SQL Design Project

Relational Database Design and SQL Implementation

**Prepared as a cleaned portfolio version of an academic
Introduction to Database group project.**

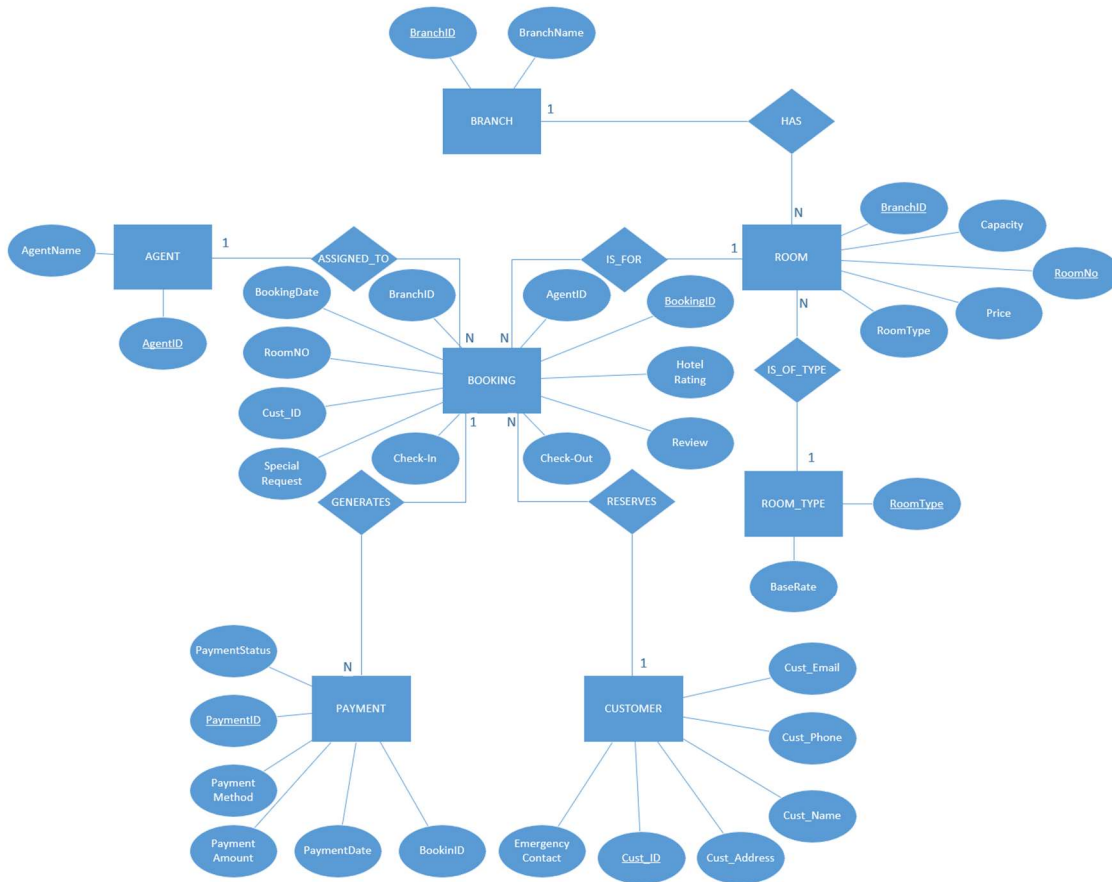
Focus Areas:

- Entity Relationship Diagram**
- Database Schema**
- SQL DDL**
- SQL DML**
- Query Result Analysis**
- Hotel Booking Data Model**

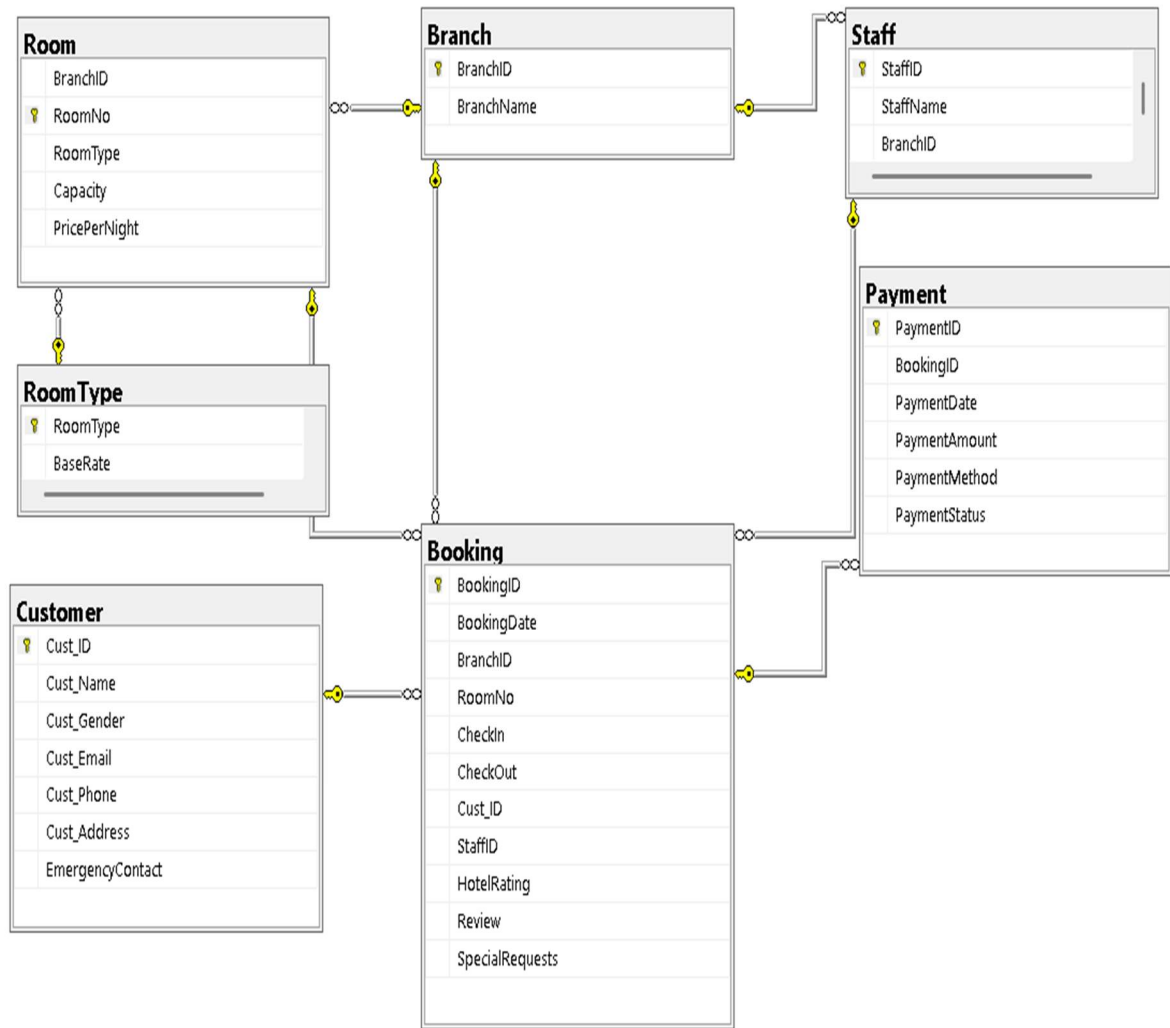
Table of Contents

Cover Page	Error! Bookmark not defined.
Table of Contents	1
Database Schema	3
Entity Relationship Diagram (ERD)	4
SQL-Data Definition Language (DDL)	5
Branch Table	5
Room Type Table	6
Room Table	7
Customer Table	9
Staff Table	11
Booking Table	13
Payment Table	15
SQL-Data Manipulation Language (DML)	17
Question 1	17
Question 2	18
Question 3	20

Database Schema



Fig(1.0) Entity-Relationship Diagram using Crow's Foot notation.



Fig(1.1) Entity Relationship Diagram generated from DBMS

SQL-Data Definition Language (DDL)

```

2 CREATE TABLE Branch (
3     BranchID      nvarchar(10) PRIMARY KEY,
4     BranchName    nvarchar(50)
5 );
6
7 INSERT INTO Branch (BranchID, BranchName)
8 VALUES
9     ('B01', 'KL Branch'),
10    ('B02', 'Penang Branch'),
11    ('B03', 'Johor Branch'),
12    ('B04', 'Melaka Branch'),
13    ('B05', 'Sabah Branch'),
14    ('B06', 'Selangor Branch'),
15    ('B07', 'Perak Branch'),
16    ('B08', 'Kedah Branch'),
17    ('B09', 'Kelantan Branch'),
18    ('B10', 'Terengganu Branch'),
19    ('B11', 'Pahang Branch'),
20    ('B12', 'Sarawak Branch'),
21    ('B13', 'Putrajaya Branch'),
22    ('B14', 'Negeri Sembilan Branch'),
23    ('B15', 'Perlis Branch'),
24    ('B16', 'Labuan Branch'),
25    ('B17', 'Langkawi Branch'),
26    ('B18', 'Ipoh Branch'),
27    ('B19', 'Kuantan Branch'),
28    ('B20', 'Kota Bharu Branch');
29
30 Select*from Branch;
31
32
33
34 CREATE TABLE RoomType(

```

74 %

Results Messages

	BranchID	BranchName
3	B03	Johor Branch
4	B04	Melaka Branch
5	B05	Sabah Branch
6	B06	Selangor Branch
7	B07	Perak Branch
8	B08	Kedah Branch
9	B09	Kelantan Branch
10	B10	Terengganu Branch
11	B11	Pahang Branch
12	B12	Sarawak Branch
13	B13	Putrajaya Branch
14	B14	Negeri Sembilan Branch
15	B15	Perlis Branch
16	B16	Labuan Branch
17	B17	Langkawi Branch
18	B18	Ipoh Branch
19	B19	Kuantan Branch
20	B20	Kota Bharu Branch

Fig(1.2) Branch Table with its query statement

```

34 CREATE TABLE RoomType(
35     RoomType      nvarchar(40) PRIMARY KEY,
36     BaseRate      decimal(6,1),
37 );
38
39 INSERT INTO RoomType (RoomType, BaseRate)
40 VALUES
41 ('Standard Double', 150),
42 ('Deluxe Double', 250),
43 ('Standard Twin', 200),
44 ('Junior Suite', 320),
45 ('Executive Suite', 400),
46 ('Family Suite', 450),
47 ('Presidential Suite', 800),
48 ('Single', 80),
49 ('Twin Deluxe', 220),
50 ('Queen', 160),
51 ('King', 200),
52 ('Studio', 140),
53 ('Apartment One-Bed', 600),
54 ('Apartment Two-Bed', 900),
55 ('Standard Triple', 210),
56 ('Business Double', 250),
57 ('Honeymoon Suite', 650),
58 ('Penthouse', 1000),
59 ('Accessible Room', 140),
60 ('Economy Double', 100);
61 select*from RoomType;
62 -----
63
64 CREATE TABLE Room (
65     BranchID      nvarchar(10),
66     RoomNo        nvarchar(10) PRIMARY KEY,

```

74 % 2 0

Results Messages

	RoomType	BaseRate
1	Accessible Room	140.0
2	Apartment One-Bed	600.0
3	Apartment Two-Bed	900.0
4	Business Double	250.0
5	Deluxe Double	250.0
6	Economy Double	100.0
7	Executive Suite	400.0
8	Family Suite	450.0
9	Honeymoon Suite	650.0
10	Junior Suite	320.0
11	King	200.0
12	Penthouse	1000.0
13	Presidential Suite	800.0
14	Queen	160.0
15	Single	80.0
16	Standard Double	150.0
17	Standard Triple	210.0
18	Standard Twin	200.0
19	Studio	140.0
20	Twin Deluxe	220.0

Fig(1.3) Room Type Table with its query statemen

	BranchID	RoomNo	RoomType	Capacity	PricePerNight
1	B01	B01-101	Standard Double	2	150.0
2	B01	B01-201	Standard Twin	2	200.0
3	B01	B01-303	Junior Suite	3	320.0
4	B02	B02-101	Deluxe Double	3	250.0
5	B02	B02-201	Queen	2	160.0
6	B02	B02-303	Executive Suite	4	400.0
7	B03	B03-101	Economy Double	2	100.0
8	B03	B03-201	Business Double	2	250.0
9	B03	B03-303	Family Suite	4	450.0
10	B04	B04-101	Accessible Room	2	140.0
11	B04	B04-201	Standard Triple	3	210.0
12	B04	B04-303	Honeymoon Suite	2	650.0
13	B05	B05-101	Studio	2	140.0
14	B05	B05-201	Twin Deluxe	2	220.0
15	B05	B05-303	King	2	200.0
16	B06	B06-101	Single	1	80.0
17	B06	B06-201	Standard Double	2	150.0
18	B06	B06-303	Executive Suite	4	400.0
19	B07	B07-101	Queen	2	160.0
20	B07	B07-201	Standard Twin	2	200.0
21	B07	B07-303	Apartment One-Bed	2	600.0
22	B08	B08-101	Economy Double	2	100.0
23	B08	B08-201	Deluxe Double	3	250.0
24	B08	B08-303	Family Suite	4	450.0
25	B09	B09-101	Accessible Room	2	140.0
26	B09	B09-201	Business Double	2	250.0
27	B09	B09-303	Apartment Two-Bed	4	900.0
28	B10	B10-101	Standard Double	2	150.0
29	B10	B10-201	Standard Triple	3	210.0
30	B10	B10-303	Presidential Suite	6	800.0
31	B11	B11-101	Studio	2	140.0
32	B11	B11-201	Twin Deluxe	2	220.0
33	B11	B11-303	King	2	200.0
34	B12	B12-101	Single	1	80.0
35	B12	B12-201	Queen	2	160.0
36	B12	B12-303	Junior Suite	3	320.0
37	B13	B13-101	Economy Double	2	100.0
38	B13	B13-201	Standard Twin	2	200.0
39	B13	B13-303	Executive Suite	4	400.0
40	B14	B14-101	Accessible Room	2	140.0
41	B14	B14-201	Business Double	2	250.0
42	B14	B14-303	Studio	2	140.0
43	B15	B15-101	Single	1	80.0
44	B15	B15-201	Queen	2	160.0
45	B15	B15-303	Junior Suite	3	320.0
46	B16	B16-101	Studio	2	140.0
47	B16	B16-201	Twin Deluxe	2	220.0
48	B16	B16-303	Apartment One-Bed	2	600.0
49	B17	B17-101	Single	1	80.0
50	B17	B17-201	Queen	2	160.0
51	B17	B17-303	Apartment Two-Bed	4	900.0
52	B18	B18-101	Economy Double	2	100.0
53	B18	B18-201	Standard Triple	3	210.0
54	B18	B18-303	Junior Suite	3	320.0
55	B19	B19-101	Accessible Room	2	140.0
56	B19	B19-201	Standard Twin	2	200.0
57	B19	B19-303	Executive Suite	4	400.0
58	B20	B20-101	Standard Double	2	150.0
59	B20	B20-201	Business Double	2	250.0
60	B20	B20-303	Penthouse	6	1000.0

Fig(1.5) Room Table Query Statement


```

140 CREATE TABLE Customer (
141     Cust_ID          nvarchar(10)    PRIMARY KEY,
142     Cust_Name        nvarchar(100),
143     Cust_Gender      nvarchar(10),
144     Cust_Email        nvarchar(100),
145     Cust_Phone        nvarchar(20),
146     Cust_Address      nvarchar(100),
147     EmergencyContact  nvarchar(20)
148 );
149
150 INSERT INTO Customer (Cust_ID, Cust_Name, Cust_Gender, Cust_Email, Cust_Phone, Cust_Address, EmergencyContact)
151 VALUES
152 ('C001', 'Ali Hassan', 'Male', 'ali@mail.com', '0123456789', 'Kuala Lumpur, MY', '0123456789'),
153 ('C002', 'Siti Rahman', 'Female', 'siti@mail.com', '0173628748', 'Penang, MY', '0122933844'),
154 ('C003', 'John Lee', 'Male', 'john@mail.com', '0122998347', 'Johor, MY', '0182275493'),
155 ('C004', 'Noor Amin', 'Male', 'noor@mail.com', '0122399874', 'Melaka, MY', '0183337777'),
156 ('C005', 'Mohd Ibrahim', 'Male', 'mohd@mail.com', '0198776253', 'Sabah, MY', '0173826893'),
157 ('C006', 'Sara Tan', 'Female', 'sara@mail.com', '0199721134', 'Selangor, MY', '0173675296'),
158 ('C007', 'Aminah Yusuf', 'Female', 'aminah@mail.com', '011223344', 'Kedah, MY', '0134455667'),
159 ('C008', 'Wei Chen', 'Male', 'wei.chen@mail.com', '019887766', 'Penang, MY', '0128877665'),
160 ('C009', 'Hassan Ali', 'Male', 'hassan@mail.com', '017554433', 'Johor Bahru, MY', '0195544332'),
161 ('C010', 'Nur Aisyah', 'Female', 'aisyah@mail.com', '010998877', 'Kuala Lumpur, MY', '0119988776'),
162 ('C011', 'Samuel Tan', 'Male', 'sam.tan@mail.com', '016336699', 'Selangor, MY', '0183366991'),
163 ('C012', 'Fatimah Noor', 'Female', 'fatimah@mail.com', '018765432', 'Melaka, MY', '0177654321'),
164 ('C013', 'Irfan Rahim', 'Male', 'irfan@mail.com', '012345120', 'Ipoh, MY', '0123400123'),
165 ('C014', 'Zara Lim', 'Female', 'zara.lim@mail.com', '011777555', 'Kuantan, MY', '0147775551'),
166 ('C015', 'Ravi Kumar', 'Male', 'ravi@mail.com', '019111222', 'Seremban, MY', '0131112223'),
167 ('C016', 'Anita Devi', 'Female', 'anita@mail.com', '015222333', 'Kota Bharu, MY', '0192223334'),
168 ('C017', 'Michael Ong', 'Male', 'm.ong@mail.com', '012909080', 'Langkawi, MY', '0179090801'),
169 ('C018', 'Nurul Izzah', 'Female', 'izzah@mail.com', '018606070', 'Labuan, MY', '0126060708'),
170 ('C019', 'Yap Siew', 'Female', 'yap.siew@mail.com', '017333222', 'Alor Setar, MY', '0163332221'),
171 ('C020', 'John Smith', 'Male', 'john.smith@mail.com', '019444111', 'Kota Bharu, MY', '0114441119');

```

Fig(1.6) Customer Table

Results		Messages					
	Cust_ID	Cust_Name	Cust_Gender	Cust_Email	Cust_Phone	Cust_Address	EmergencyContact
1	C001	Ali Hassan	Male	ali@mail.com	0123456789	Kuala Lumpur, MY	0123456789
2	C002	Siti Rahman	Female	siti@mail.com	0173628748	Penang, MY	0122933844
3	C003	John Lee	Male	john@mail.com	0122998347	Johor, MY	0182275493
4	C004	Noor Amin	Male	noor@mail.com	0122399874	Melaka, MY	0183337777
5	C005	Mohd Ibrahim	Male	mohd@mail.com	0198776253	Sabah, MY	0173826893
6	C006	Sara Tan	Female	sara@mail.com	0199721134	Selangor, MY	0173675296
7	C007	Aminah Yusuf	Female	aminah@mail.com	011223344	Kedah, MY	0134455667
8	C008	Wei Chen	Male	wei.chen@mail.com	019887766	Penang, MY	0128877665
9	C009	Hassan Ali	Male	hassan@mail.com	017554433	Johor Bahru, MY	0195544332
10	C010	Nur Aisyah	Female	aisyah@mail.com	010998877	Kuala Lumpur, MY	0119988776
11	C011	Samuel Tan	Male	sam.tan@mail.com	016336699	Selangor, MY	0183366991
12	C012	Fatimah Noor	Female	fatimah@mail.com	018765432	Melaka, MY	0177654321
13	C013	Irfan Rahim	Male	irfan@mail.com	012345120	Ipoh, MY	0123400123
14	C014	Zara Lim	Female	zara.lim@mail.com	011777555	Kuantan, MY	0147775551
15	C015	Ravi Kumar	Male	ravi@mail.com	019111222	Seremban, MY	0131112223
16	C016	Anita Devi	Female	anita@mail.com	015222333	Kota Bharu, MY	0192223334
17	C017	Michael Ong	Male	m.ong@mail.com	012909080	Langkawi, MY	0179090801
18	C018	Nurul Izzah	Female	izzah@mail.com	018606070	Labuan, MY	0126060708
19	C019	Yap Siew	Female	yap.siew@mail.com	017333222	Alor Setar, MY	0163332221
20	C020	John Smith	Male	john.smith@mail.c...	019444111	Kota Bharu, MY	0114441119

Fig(1.7) Customer Table query statement

```

178 CREATE TABLE Staff(
179     StaffID    nvarchar(10) PRIMARY KEY,
180     StaffName  nvarchar(50),
181     BranchID  nvarchar(10),
182     FOREIGN KEY (BranchID) REFERENCES Branch(BranchID)
183 );
184
185
186 INSERT INTO Staff (StaffID, StaffName, BranchID) VALUES
187 ('G01', 'Mr.Ahmad', 'B01'), ('G02', 'Mr.Mohammad', 'B01'),
188 ('G03', 'Ms.Farah', 'B02'), ('G04', 'Mr.Lim', 'B02'),
189 ('G05', 'Ms.Nora', 'B03'), ('G06', 'Mr.Zain', 'B03'),
190 ('G07', 'Ms.Chia', 'B04'), ('G08', 'Mr.Razak', 'B04'),
191 ('G09', 'Ms.Huda', 'B05'), ('G10', 'Mr.Faiz', 'B05'),
192 ('G11', 'Ms.Nadia', 'B06'), ('G12', 'Mr.Tan', 'B06'),
193 ('G13', 'Ms.Azura', 'B07'), ('G14', 'Mr.Hakim', 'B07'),
194 ('G15', 'Ms.Lee', 'B08'), ('G16', 'Mr.Ray', 'B08'),
195 ('G17', 'Ms.Sophia', 'B09'), ('G18', 'Mr.Kamarul', 'B09'),
196 ('G19', 'Ms.Jenny', 'B10'), ('G20', 'Mr.Arif', 'B10'),
197 ('G21', 'Ms.Aisha', 'B11'), ('G22', 'Mr.Hafiz', 'B11'),
198 ('G23', 'Ms.Lina', 'B12'), ('G24', 'Mr.Daniel', 'B12'),
199 ('G25', 'Ms.Sharifah', 'B13'), ('G26', 'Mr.Farid', 'B13'),
200 ('G27', 'Ms.Raina', 'B14'), ('G28', 'Mr.Hadi', 'B14'),
201 ('G29', 'Ms.Kylie', 'B15'), ('G30', 'Mr.Amir', 'B15'),
202 ('G31', 'Ms.Fiona', 'B16'), ('G32', 'Mr.Yusuf', 'B16'),
203 ('G33', 'Ms.Joan', 'B17'), ('G34', 'Mr.Akmal', 'B17'),
204 ('G35', 'Ms.Tan', 'B18'), ('G36', 'Mr.Rizal', 'B18'),
205 ('G37', 'Ms.Erin', 'B19'), ('G38', 'Mr.Zul', 'B19'),
206 ('G39', 'Ms.Julie', 'B20'), ('G40', 'Mr.Nazmi', 'B20');

```

Fig(1.8) Staff Table

	StaffID	StaffName	BranchID
1	G01	Mr.Ahmad	B01
2	G02	Mr.Mohammad	B01
3	G03	Ms.Farah	B02
4	G04	Mr.Lim	B02
5	G05	Ms.Nora	B03
6	G06	Mr.Zain	B03
7	G07	Ms.Chia	B04
8	G08	Mr.Razak	B04
9	G09	Ms.Huda	B05
10	G10	Mr.Faiz	B05
11	G11	Ms.Nadia	B06
12	G12	Mr.Tan	B06
13	G13	Ms.Azura	B07
14	G14	Mr.Hakim	B07
15	G15	Ms.Lee	B08
16	G16	Mr.Ray	B08
17	G17	Ms.Sophia	B09
18	G18	Mr.Kamarul	B09
19	G19	Ms.Jenny	B10
20	G20	Mr.Arif	B10
21	G21	Ms.Aisha	B11
22	G22	Mr.Hafiz	B11
23	G23	Ms.Lina	B12
24	G24	Mr.Daniel	B12
25	G25	Ms.Sharifah	B13
26	G26	Mr.Farid	B13
27	G27	Ms.Raina	B14
28	G28	Mr.Hadi	B14
29	G29	Ms.Kylie	B15
30	G30	Mr.Amir	B15
31	G31	Ms.Fiona	B16
32	G32	Mr.Yusuf	B16
33	G33	Ms.Joan	B17
34	G34	Mr.Akmal	B17
35	G35	Ms.Tan	B18
36	G36	Mr.Rizal	B18
37	G37	Ms.Erin	B19
38	G38	Mr.Zul	B19
39	G39	Ms.Julie	B20
40	G40	Mr.Nazmi	B20

Fig(1.9) Staff Table query statement


```

CREATE TABLE Booking(
    BookingID          nvarchar(10) PRIMARY KEY,
    BookingDate        date,
    BranchID           nvarchar(10),
    RoomNo             nvarchar(10),
    CheckIn            date,
    CheckOut           date,
    Cust_ID            nvarchar(10),
    StaffID            nvarchar(10),
    HotelRating        int,
    Review             nvarchar(300),
    SpecialRequests    nvarchar(300),
    foreign key (RoomNo) references Room(RoomNo),
    foreign key (BranchID) references Branch(BranchID),
    foreign key (Cust_ID) references Customer(Cust_ID),
    foreign key (StaffID) references Staff(StaffID),
);

INSERT INTO Booking
(BookingID, BookingDate, BranchID, RoomNo, CheckIn, CheckOut, Cust_ID, StaffID, HotelRating, Review, SpecialRequests)
VALUES
('BK001', '2024-01-05', 'B01', 'B01-101', '2024-01-12', '2024-01-12', 'C001', 'G01', 5, 'Excellent Service', 'Non-smoking, Late check-in'),
('BK007', '2024-02-10', 'B01', 'B01-201', '2024-02-14', '2024-02-14', 'C007', 'G02', 4, 'Good location', NULL),
('BK019', '2024-10-10', 'B01', 'B01-303', '2024-10-18', '2024-10-18', 'C019', 'G01', 3, 'Not bad overall', 'Late check-in'),
('BK002', '2024-03-12', 'B02', 'B02-101', '2024-03-23', '2024-03-23', 'C002', 'G03', 4, 'Nice, but noisy near the pool', 'Near Pool'),
('BK008', '2024-02-18', 'B02', 'B02-201', '2024-02-22', '2024-02-22', 'C008', 'G03', 5, 'Great breakfast', 'Near elevator'),
('BK018', '2024-09-03', 'B02', 'B02-303', '2024-09-10', '2024-09-10', 'C018', 'G03', 5, 'Executive suite was excellent', 'Early check-in'),
('BK003', '2024-06-02', 'B03', 'B03-101', '2024-06-12', '2024-06-12', 'C003', 'G04', 3, 'Average experience', 'Early Check-in'),
('BK009', '2024-03-05', 'B03', 'B03-201', '2024-03-12', '2024-03-12', 'C009', 'G04', 3, 'Average stay', 'Quiet room'),
('BK021', '2024-11-01', 'B03', 'B03-303', '2024-11-08', '2024-11-08', 'C009', 'G06', 4, 'Family room was spacious', 'Extra towels'),
('BK004', '2024-08-15', 'B04', 'B04-101', '2024-08-20', '2024-08-20', 'C004', 'G05', 5, 'Very Satisfied, Clean room', 'Non-Smoking, Extra pillows'),
('BK010', '2024-04-01', 'B04', 'B04-201', '2024-04-08', '2024-04-08', 'C010', 'G05', 5, 'Loved the suite', NULL),
('BK020', '2024-11-11', 'B04', 'B04-303', '2024-11-20', '2024-11-20', 'C020', 'G05', 4, NULL, NULL),
('BK005', '2024-09-10', 'B05', 'B05-101', '2024-09-15', '2024-09-15', 'C005', 'G06', 4, 'Comfortable stay', 'Extra bed'),
('BK011', '2024-04-15', 'B05', 'B05-201', '2024-04-20', '2024-04-20', 'C011', 'G06', 1, 'Too much noise!', 'Extra pillows'),
('BK006', '2024-10-01', 'B06', 'B06-101', '2024-10-08', '2024-10-08', 'C006', 'G07', 3, 'Excellent, Will come again', 'Late check-out, Non-Smoking'),
('BK012', '2024-05-02', 'B06', 'B06-201', '2024-05-10', '2024-05-10', 'C012', 'G07', 3, 'Decent, value for money', 'Non-smoking'),
('BK013', '2024-05-21', 'B07', 'B07-201', '2024-05-28', '2024-05-28', 'C013', 'G08', 4, 'Nice staff', 'Near lobby'),
('BK014', '2024-06-03', 'B08', 'B08-201', '2024-06-09', '2024-06-09', 'C014', 'G09', 5, 'Fantastic service', 'Sea view'),
('BK015', '2024-07-07', 'B09', 'B09-201', '2024-07-15', '2024-07-15', 'C015', 'G10', 2, 'Bad attitude from the reception', NULL),
('BK016', '2024-07-25', 'B10', 'B10-201', '2024-08-01', '2024-08-01', 'C016', 'G11', 5, 'Perfect for family', 'Baby cot'),
('BK017', '2024-08-05', 'B11', 'B11-201', '2024-08-12', '2024-08-12', 'C017', 'G12', 4, NULL, 'Near exit'),
('BK022', '2024-02-15', 'B12', 'B12-201', '2024-02-20', '2024-02-20', 'C012', 'G23', 5, 'Wonderful long stay', 'Extra pillows'),
('BK023', '2024-03-11', 'B13', 'B13-201', '2024-03-18', '2024-03-18', 'C013', 'G25', 3, 'Okay stay, clean facilities', 'Quiet room'),
('BK024', '2024-04-07', 'B14', 'B14-201', '2024-04-12', '2024-04-12', 'C014', 'G27', 4, 'Modern design, friendly staff', 'Non-smoking'),
('BK025', '2024-05-18', 'B15', 'B15-201', '2024-05-22', '2024-05-22', 'C015', 'G29', 3, 'Small but comfy', 'Near elevator'),
('BK026', '2024-06-14', 'B16', 'B16-201', '2024-06-19', '2024-06-19', 'C016', 'G31', 5, 'Great sea view', 'Sea view'),
('BK027', '2024-07-03', 'B17', 'B17-201', '2024-07-09', '2024-07-09', 'C017', 'G33', 4, 'Relaxing stay on the island', 'Late check-out'),
('BK028', '2024-08-11', 'B18', 'B18-201', '2024-08-18', '2024-08-18', 'C018', 'G35', 5, 'Perfect for family trip', NULL),
('BK029', '2024-09-07', 'B19', 'B19-201', '2024-09-15', '2024-09-15', 'C019', 'G37', 4, 'Lovely coastal hotel', 'Extra bed'),
('BK030', '2024-10-14', 'B20', 'B20-201', '2024-10-20', '2024-10-20', 'C020', 'G39', 5, 'Very welcoming staff', 'Non-smoking');

```

Fig(1.10) Booking Table with its query

	BookingID	BookingDate	BranchID	RoomNo	CheckIn	CheckOut	Cust_ID	StaffID	HotelRating	Review	SpecialRequests
1	BK001	2024-01-05	B01	B01-101	2024-01-12	2024-01-12	C001	G01	5	Excellent Service	Non-smoking, Late check-in
2	BK002	2024-03-12	B02	B02-101	2024-03-23	2024-03-23	C002	G03	4	Nice, but noisy near the pool	Near Pool
3	BK003	2024-06-02	B03	B03-101	2024-06-12	2024-06-12	C003	G04	3	Average experience	Early Check-in
4	BK004	2024-08-15	B04	B04-101	2024-08-20	2024-08-20	C004	G05	5	Very Satisfied, Clean room	Non-Smoking, Extra pillows
5	BK005	2024-09-10	B05	B05-101	2024-09-15	2024-09-15	C005	G06	4	Comfortable stay	Extra bed
6	BK006	2024-10-01	B06	B06-101	2024-10-08	2024-10-08	C006	G07	3	Excellent, Will come again	Late check-out, Non-Smoking
7	BK007	2024-02-10	B01	B01-201	2024-02-14	2024-02-14	C007	G02	4	Good location	NULL
8	BK008	2024-02-18	B02	B02-201	2024-02-22	2024-02-22	C008	G03	5	Great breakfast	Near elevator
9	BK009	2024-03-05	B03	B03-201	2024-03-12	2024-03-12	C009	G04	3	Average stay	Quiet room
10	BK010	2024-04-01	B04	B04-201	2024-04-08	2024-04-08	C010	G05	5	Loved the suite	NULL
11	BK011	2024-04-15	B05	B05-201	2024-04-20	2024-04-20	C011	G06	1	Too much noise!	Extra pillows
12	BK012	2024-05-02	B06	B06-201	2024-05-10	2024-05-10	C012	G07	3	Decent, value for money	Non-smoking
13	BK013	2024-05-21	B07	B07-201	2024-05-28	2024-05-28	C013	G08	4	Nice staff	Near lobby
14	BK014	2024-06-03	B08	B08-201	2024-06-09	2024-06-09	C014	G09	5	Fantastic service	Sea view
15	BK015	2024-07-07	B09	B09-201	2024-07-15	2024-07-15	C015	G10	2	Bad attitude from the reception	NULL
16	BK016	2024-07-25	B10	B10-201	2024-08-01	2024-08-01	C016	G11	5	Perfect for family	Baby cot
17	BK017	2024-08-05	B11	B11-201	2024-08-12	2024-08-12	C017	G12	4	NULL	Near exit
18	BK018	2024-09-03	B02	B02-303	2024-09-10	2024-09-10	C018	G03	5	Executive suite was excellent	Early check-in
19	BK019	2024-10-10	B01	B01-303	2024-10-18	2024-10-18	C019	G01	3	Not bad overall	Late check-in
20	BK020	2024-11-11	B04	B04-303	2024-11-20	2024-11-20	C020	G05	4	NULL	NULL
21	BK021	2024-11-01	B03	B03-303	2024-11-08	2024-11-08	C009	G06	4	Family room was spacious	Extra towels
22	BK022	2024-02-15	B12	B12-201	2024-02-20	2024-02-20	C012	G23	5	Wonderful long stay	Extra pillows
23	BK023	2024-03-11	B13	B13-201	2024-03-18	2024-03-18	C013	G25	3	Okay stay, clean facilities	Quiet room
24	BK024	2024-04-07	B14	B14-201	2024-04-12	2024-04-12	C014	G27	4	Modern design, friendly staff	Non-smoking
25	BK025	2024-05-18	B15	B15-201	2024-05-22	2024-05-22	C015	G29	3	Small but comfy	Near elevator
26	BK026	2024-06-14	B16	B16-201	2024-06-19	2024-06-19	C016	G31	5	Great sea view	Sea view
27	BK027	2024-07-03	B17	B17-201	2024-07-09	2024-07-09	C017	G33	4	Relaxing stay on the island	Late check-out
28	BK028	2024-08-11	B18	B18-201	2024-08-18	2024-08-18	C018	G35	5	Perfect for family trip	NULL
29	BK029	2024-09-07	B19	B19-201	2024-09-15	2024-09-15	C019	G37	4	Lovely coastal hotel	Extra bed
30	BK030	2024-10-14	B20	B20-201	2024-10-20	2024-10-20	C020	G39	5	Very welcoming staff	Non-smoking

Fig(1.11) Booking Table with its query statemen

```

268 CREATE TABLE Payment(
269     PaymentID      nvarchar(10) PRIMARY KEY,
270     BookingID       nvarchar(10),
271     PaymentDate     date,
272     PaymentAmount   decimal(4,1),
273     PaymentMethod   nvarchar(50),
274     PaymentStatus   nvarchar(50),
275     foreign key (BookingID) references Booking(BookingID)
276 );
277
278 insert into Payment (PaymentID, BookingID, PaymentDate, PaymentAmount, PaymentMethod, PaymentStatus)
279 VALUES
280 ('PAY001', 'BK001', '2024-01-05', 450, 'MasterCard', 'Fully Paid'),
281 ('PAY002', 'BK001', '2024-01-12', 450, 'Cash', 'Paid by a company'),
282 ('PAY003', 'BK002', '2024-03-23', 540, 'Visa', 'Fully Paid'),
283 ('PAY004', 'BK003', NULL, 500, NULL, 'Not Paid Yet'),
284 ('PAY005', 'BK004', '2024-08-20', 300, 'Cash', 'Fully Paid'),
285 ('PAY006', 'BK005', '2024-09-15', 360, 'MasterCard', 'Fully Paid'),
286 ('PAY007', 'BK006', '2024-10-01', 750, 'MasterCard', 'Fully Paid'),
287 ('PAY008', 'BK007', '2024-02-10', 300, 'Visa', 'Deposit Paid'),
288 ('PAY009', 'BK008', '2024-02-22', 320, 'MasterCard', 'Fully Paid'),
289 ('PAY010', 'BK009', NULL, 200, NULL, 'Not Paid Yet'),
290 ('PAY011', 'BK010', '2024-04-01', 500, 'Visa', 'Paid by a company'),
291 ('PAY012', 'BK011', '2024-04-20', 280, 'Cash', 'Fully Paid'),
292 ('PAY013', 'BK012', NULL, 180, NULL, 'Not Paid Yet'),
293 ('PAY014', 'BK013', '2024-05-28', 220, 'MasterCard', 'Fully Paid'),
294 ('PAY015', 'BK014', '2024-06-09', 250, 'Visa', 'Fully Paid'),
295 ('PAY016', 'BK015', '2024-07-15', 180, 'Cash', 'Fully Paid'),
296 ('PAY017', 'BK016', '2024-08-01', 450, 'MasterCard', 'Fully Paid'),
297 ('PAY018', 'BK017', '2024-08-12', 150, 'Visa', 'Fully Paid'),
298 ('PAY019', 'BK018', '2024-09-10', 400, 'MasterCard', 'Fully Paid'),
299 ('PAY020', 'BK019', NULL, 250, NULL, 'Not Paid Yet'),
300 ('PAY021', 'BK020', '2024-11-20', 350, 'Cash', 'Fully Paid'),
301 ('PAY022', 'BK021', '2024-11-08', 480, 'Visa', 'Fully Paid'),
302 ('PAY023', 'BK022', '2024-02-20', 600, 'MasterCard', 'Fully Paid'),
303 ('PAY024', 'BK023', NULL, 300, NULL, 'Not Paid Yet'),
304 ('PAY025', 'BK024', '2024-04-12', 420, 'Cash', 'Fully Paid'),
305 ('PAY026', 'BK025', '2024-05-22', 200, 'Visa', 'Fully Paid'),
306 ('PAY027', 'BK026', '2024-06-19', 500, 'MasterCard', 'Fully Paid'),
307 ('PAY028', 'BK027', '2024-07-09', 320, 'Cash', 'Fully Paid'),
308 ('PAY029', 'BK028', '2024-08-18', 700, 'Visa', 'Fully Paid'),
309 ('PAY030', 'BK029', '2024-09-15', 380, 'MasterCard', 'Fully Paid'),
310 ('PAY031', 'BK030', '2024-10-20', 450, 'Cash', 'Fully Paid');
311 select*from Payment;

```

Fig(1.12) Payment Table with its query statemen

Results Messages						
	PaymentID	BookingID	PaymentDate	PaymentAmount	PaymentMethod	PaymentStatus
1	PAY001	BK001	2024-01-05	450.0	MasterCard	Fully Paid
2	PAY002	BK001	2024-01-12	450.0	Cash	Paid by a co...
3	PAY003	BK002	2024-03-23	540.0	Visa	Fully Paid
4	PAY004	BK003	NULL	500.0	NULL	Not Paid Yet
5	PAY005	BK004	2024-08-20	300.0	Cash	Fully Paid
6	PAY006	BK005	2024-09-15	360.0	MasterCard	Fully Paid
7	PAY007	BK006	2024-10-01	750.0	MasterCard	Fully Paid
8	PAY008	BK007	2024-02-10	300.0	Visa	Deposit Paid
9	PAY009	BK008	2024-02-22	320.0	MasterCard	Fully Paid
10	PAY010	BK009	NULL	200.0	NULL	Not Paid Yet
11	PAY011	BK010	2024-04-01	500.0	Visa	Paid by a co...
12	PAY012	BK011	2024-04-20	280.0	Cash	Fully Paid
13	PAY013	BK012	NULL	180.0	NULL	Not Paid Yet
14	PAY014	BK013	2024-05-28	220.0	MasterCard	Fully Paid
15	PAY015	BK014	2024-06-09	250.0	Visa	Fully Paid
16	PAY016	BK015	2024-07-15	180.0	Cash	Fully Paid
17	PAY017	BK016	2024-08-01	450.0	MasterCard	Fully Paid
18	PAY018	BK017	2024-08-12	150.0	Visa	Fully Paid
19	PAY019	BK018	2024-09-10	400.0	MasterCard	Fully Paid
20	PAY020	BK019	NULL	250.0	NULL	Not Paid Yet
21	PAY021	BK020	2024-11-20	350.0	Cash	Fully Paid
22	PAY022	BK021	2024-11-08	480.0	Visa	Fully Paid
23	PAY023	BK022	2024-02-20	600.0	MasterCard	Fully Paid
24	PAY024	BK023	NULL	300.0	NULL	Not Paid Yet
25	PAY025	BK024	2024-04-12	420.0	Cash	Fully Paid
26	PAY026	BK025	2024-05-22	200.0	Visa	Fully Paid
27	PAY027	BK026	2024-06-19	500.0	MasterCard	Fully Paid
28	PAY028	BK027	2024-07-09	320.0	Cash	Fully Paid
29	PAY029	BK028	2024-08-18	700.0	Visa	Fully Paid
30	PAY030	BK029	2024-09-15	380.0	MasterCard	Fully Paid
31	PAY031	BK030	2024-10-20	450.0	Cash	Fully Paid

Fig(1.13) Payment Table statemen

SQL-Data Manipulation Language (DML)

Question 1

" Display the customers who have not made any reservation since the beginning until the end of 2024. The result should include first name, last name, email address, and contact number."

Query

```
1--Display the customers who have not made any reservation since the beginning until the end of 2024. The result should include first name, last name, email
✓ Select Cust_Name, Cust_Email, Cust_Address, Cust_Phone from Customer
left join Booking on Customer.Cust_ID = Booking.Cust_ID
and Booking.BookingDate between '2024-01-01' and '2024-12-31' where Booking.BookingID is null
```

Fig(2.1.1)

Result

Cust_Name	Cust_Email	Cust_Address	Cust_Phone
-----------	------------	--------------	------------

Fig(2.1.2)

Explanation

The query retrieves the customers who made zero bookings in 2024. This is achieved through a **LEFT JOIN of the Customer and Booking** tables to include every customer regardless if they made a booking or not. By adding the condition that

Booking.BookingID is NULL

It effectively removes those customers who have made at least one booking, rendering only those who have made zero in that time frame.

The result set fields include the first name, last name, email address, and contact number for each customer, effectively proving they have zero reservations made between January 1 and December 31, 2024.

The result is empty since every customer has at least one booking in 2024.

Question 2

“Display the monthly revenue of the hotel by state for the year 2024 and sort the results in descending order by monthly revenue.”

Query

```
--2 Display the monthly revenue of the hotel by state for the year 2024 and sort the results in descending order by monthly revenue.
select Branch.BranchName, Payment.PaymentDate, sum(Payment.PaymentAmount) as 'Monthly Revenue' from Payment
left join Booking on Payment.BookingID = Booking.BookingID
join Branch on Booking.BranchID = Branch.BranchID
where Payment.PaymentDate between '2024-01-01' and '2024-12-31' group by Branch.BranchName, Payment.PaymentDate order by sum(Payment.PaymentAmount) desc
```

Fig(2.2.1)

Result

	BranchName	PaymentDate	Monthly Revenue
1	Selangor Branch	2024-10-01	750.0
2	Ipoh Branch	2024-08-18	700.0
3	Sarawak Branch	2024-02-20	600.0
4	Penang Branch	2024-03-23	540.0
5	Melaka Branch	2024-04-01	500.0
6	Labuan Branch	2024-06-19	500.0
7	Johor Branch	2024-11-08	480.0
8	Kota Bharu Branch	2024-10-20	450.0
9	Terengganu Branch	2024-08-01	450.0
10	KL Branch	2024-01-05	450.0
11	KL Branch	2024-01-12	450.0
12	Negeri Sembilan Branch	2024-04-12	420.0
13	Penang Branch	2024-09-10	400.0
14	Kuantan Branch	2024-09-15	380.0
15	Sabah Branch	2024-09-15	360.0
16	Melaka Branch	2024-11-20	350.0
17	Penang Branch	2024-02-22	320.0
18	Langkawi Branch	2024-07-09	320.0
19	Melaka Branch	2024-08-20	300.0
20	KL Branch	2024-02-10	300.0
21	Sabah Branch	2024-04-20	280.0
22	Kedah Branch	2024-06-09	250.0
23	Perak Branch	2024-05-28	220.0
24	Perlis Branch	2024-05-22	200.0
25	Kelantan Branch	2024-07-15	180.0
26	Pahang Branch	2024-08-12	150.0

Fig(2.2.2)

Explanation

This SQL query retrieves the total revenue received per hotel branch (in **2024**) — sorted highest to lowest. It links three tables: **Payment**, **Booking** and **Branch** to identify which **Payment** is associated with which **Branch**. The **WHERE** statement in the middle limits the query to **Payment** amounts that fall within the dates of January 1, 2024 through December 31, 2024, since we want a summary of total payments during the calendar year. In addition, **sum(PaymentAmount)** sums the **PaymentAmounts** for each branch and each payment date. Thus, **GROUP BY Branch.BranchName, Payment.PaymentDate** combines the results and **ORDER BY SUM(Payment.PaymentAmount) DESC** displays them in a ranking format so that highest revenue generating hotels are listed first.

Question 3

“Display customers who have made more than 3 reservations in the year of 2024.”

Query

```
--3 Display customers who have made more than 3 reservations in the year of 2024.
select Cust_ID, count(BookingID) as 'Customer with more than 3 reservations' from Booking
where BookingDate between '2024-01-01' AND '2024-12-31' group by Cust_ID
having count(BookingID) > 3;
```

Fig(2.3.1)

Result

Cust_ID	Customer with more than 3 reservations
---------	--

Fig(2.3.2)

Explanation

This query selects customers that made **greater than 3 bookings** in the year **2024**. It pulls the **Cust_ID** from the **Booking** table and utilizes the **COUNT(BookingID)** function to determine frequency of repeat **Cust_ID** entries. The **WHERE** date falls between January 1, 2024 and December 31, 2024. The

GROUP BY Cust_ID

makes sense to compile results and the

HAVING COUNT(BookingID) > 3

limits the results to only those customers who made greater than 3 reservations.

The result is empty since no customers made more than 3 reservations in 2024 in the database.